Synoptic LE: Soil Fluxes 2023

Roberta Peixoto

2026-05-05

Table of contents

1 Read in and collate all the files in the Processed Data folder

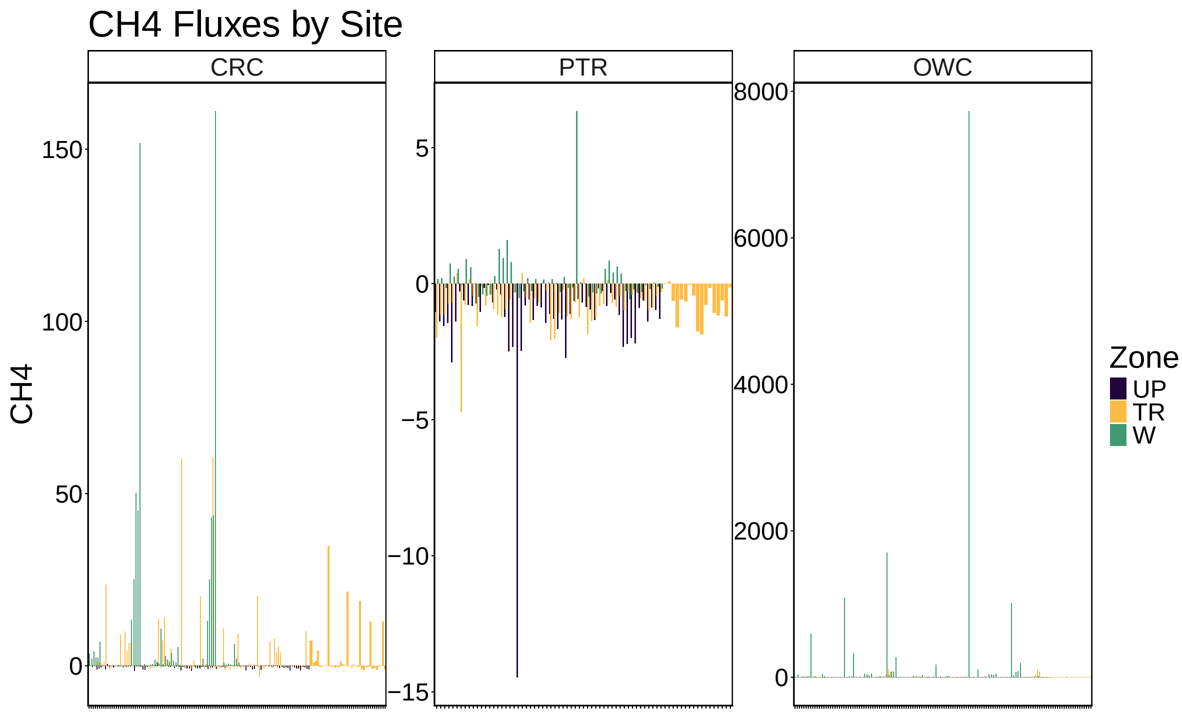
2 Remove bad fluxes and calculate how many there were

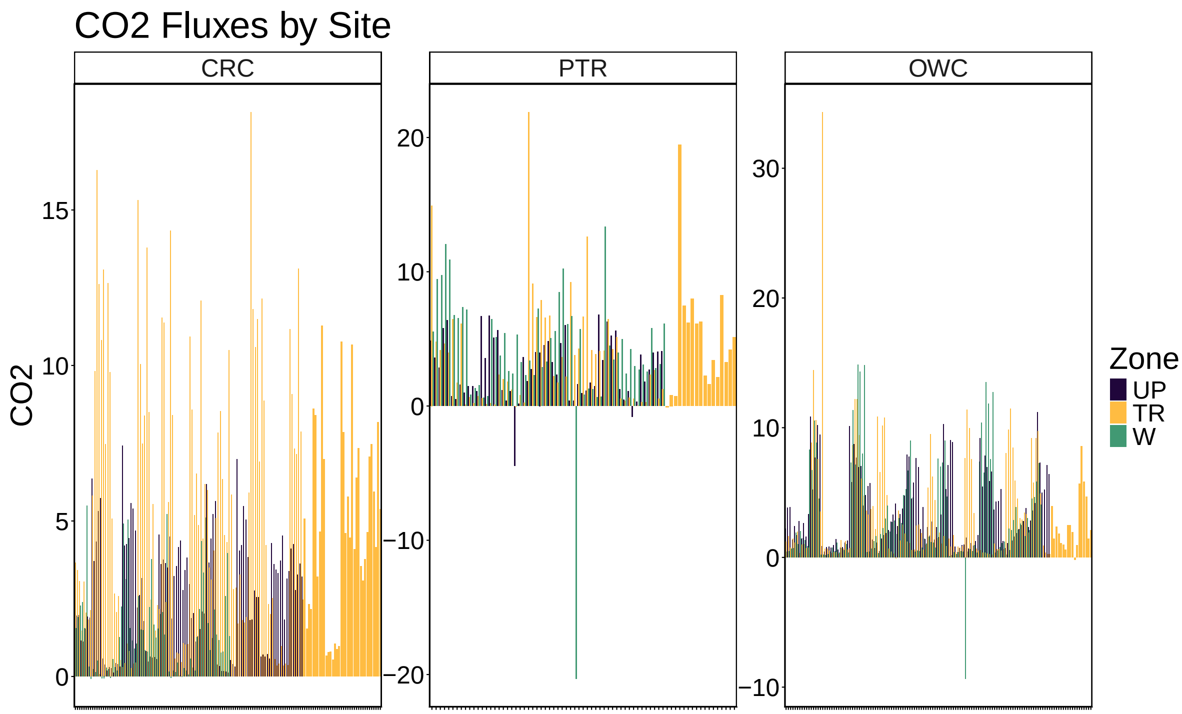
```
##Remove any samples with "remove" in the qaqc_flag columns
final_dat <- all_dat %>%
  filter(!grepl("remove", ch4_qaqc_flag, ignore.case = TRUE)) %>%
  filter(!grepl("remove", co2_qaqc_flag, ignore.case = TRUE))

##Removed fluxes in the CH4 and CO2 flux processing Rmd code, removed both CH4 and CO2 obs wh
```

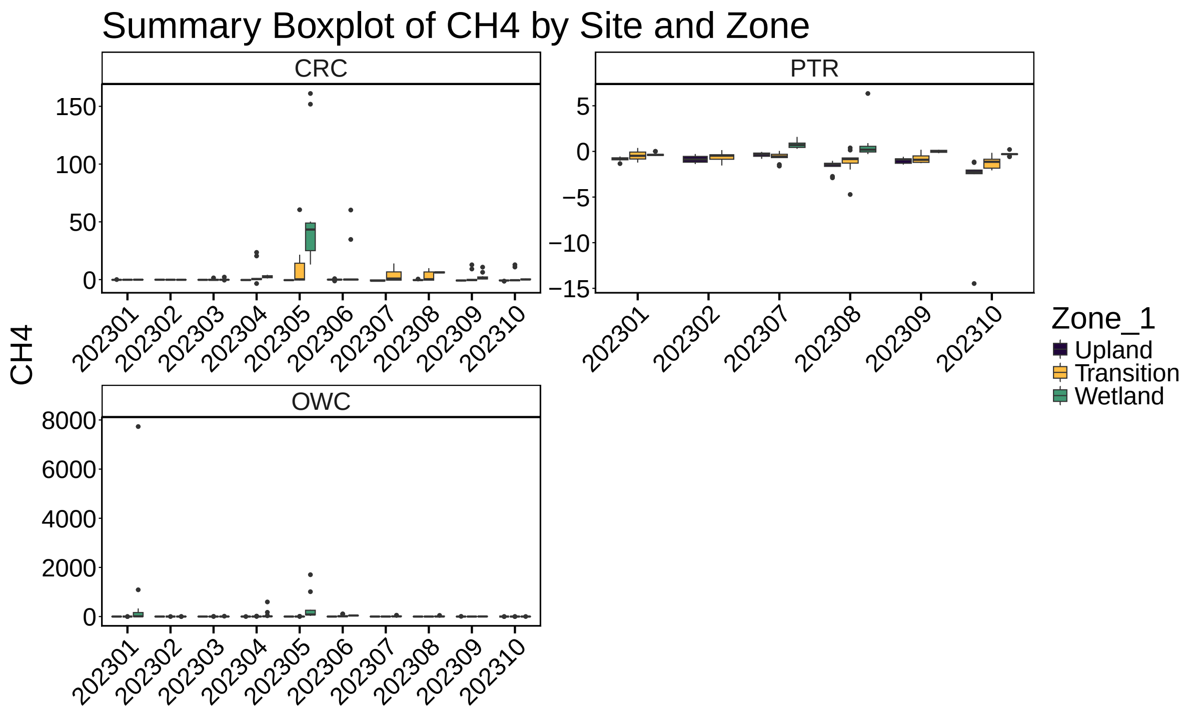
3 Write out files

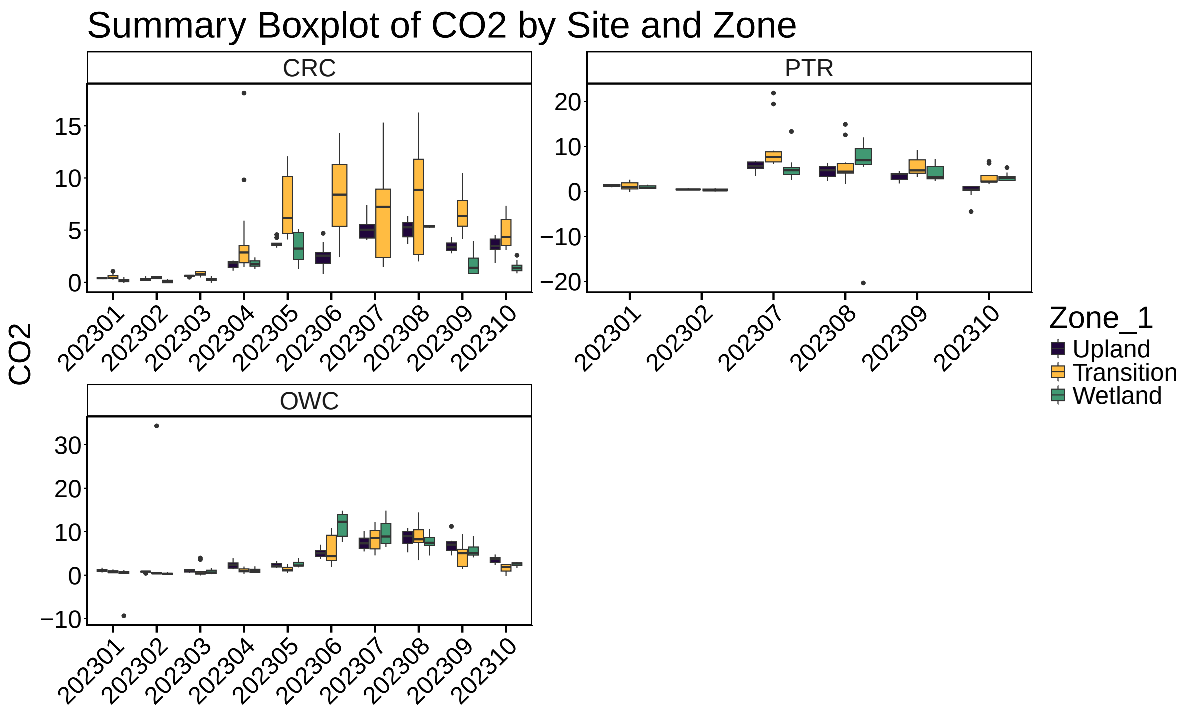
4 All Data by Year, Site and Zone



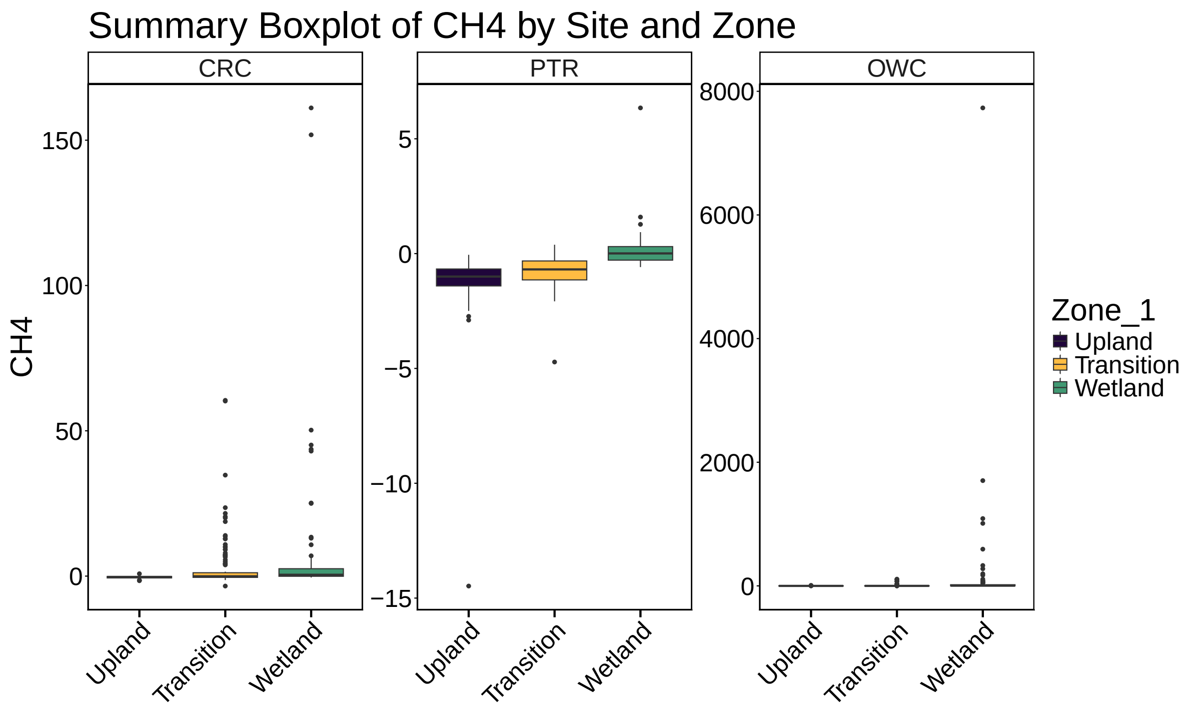

CO2 Fluxes by Site
CRC
PTR
OWC
CO2
Zone
UP
TR
W
0
5
10
15
20
10
0
−10
−20
30
20
10
0
−10

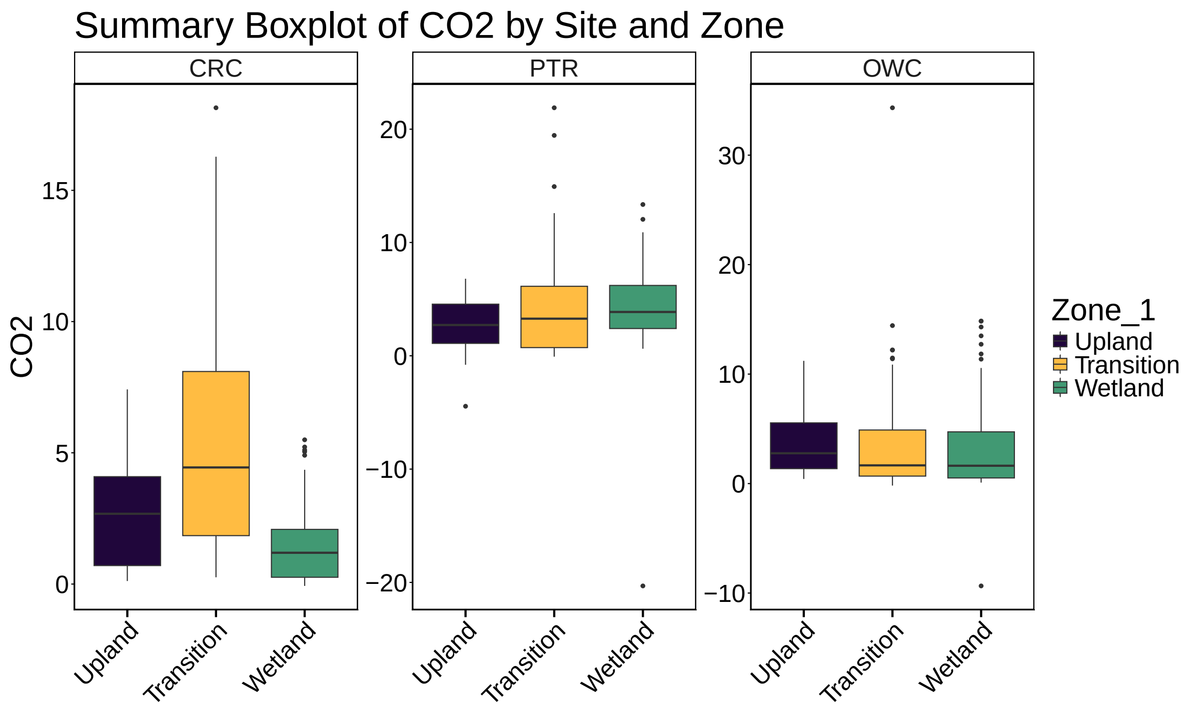
5 Boxplot summary by Date




Summary Boxplot of CO2 by Site and Zone
CRC
PTR
OWC
CO2
15
10
5
0
20
10
0
−10
−20
30
20
10
0
−10
202301
202302
202303
202304
202305
202306
202307
202308
202309
202310
Zone_1
Upland
Transition
Wetland

6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
CRC	Upland	-1.5871720	0.8138948	-0.4010641	0.1157130	7.415984	2.654480
CRC	Transition	-3.4188587	60.4782950	2.9587877	0.2588368	18.145495	5.283785
CRC	Wetland	-0.4748040	161.1104257	8.8630706	-0.0708765	5.496121	1.439939
PTR	Upland	-14.4754777	-0.0516143	-1.3680544	-4.4526614	6.802548	2.802321
PTR	Transition	-4.7196955	0.3874644	-0.7930222	-0.0758257	21.888915	3.965143
PTR	Wetland	-0.5863126	6.3454046	0.1936119	-20.3095271	13.352324	4.185905
OWC	Upland	-1.0716442	8.2741057	-0.2366732	0.4206938	11.209699	3.726660
OWC	Transition	-1.4243664	109.0540436	4.3580981	-0.1825617	34.326671	3.526997
OWC	Wetland	-2.6030380	7731.4754499	128.2821513	-9.3482966	14.846241	3.231102